

M33 — Character names + voice test + engage on the Per-Patient Console

Phase: Phase 7 follow-on (post-M32, operator-feedback fix) **Status:** DONE **Blocked by:** M30 (voice picker card scaffold), M31 (per-row persona multi-select) **Blocks:** none **Estimated effort:** 0.5 day

1. Purpose

Operator feedback on the Per-Patient Console:

"On the individual encounter control pages, use the character names rather than the character code on the pull down for the selecting patient voices. Have a button to check voice sound. Also list all the characters for the scenario so that the instructor can engage the character from the control page for the encounter control page."

Three changes, all on the voice card (#card-voice):

1. **Display names instead of IDs.** Each row now reads Mr . Hayes (with a small role tag like Hyperactive Delirium and the raw P-014 shown as a faint monospace pid below the name) — not the bare P-014 it showed before.
2. **■ Test button.** A per-row preview button synthesizes a sample phrase ("Hello, I'm Mr. Hayes. This is what my voice sounds like in the simulation.") via the selected ElevenLabs voice and plays it in the browser. Rows with no voice picked use browser SpeechSynthesis as a fallback so the instructor at least hears the default voice.
3. **■ Engage button.** A per-row link opens `/join?code={join_code}` in a popup tab so the instructor can pick that character and chat as them — turning the encounter console into the launchpad for instructor-driven character role-play. The card's heading is renamed from `"■ Character voices"` to `"■ Characters · voices · engage"` to reflect the three-job role.

The card is now a single source-of-truth for the persona roster on this encounter: list, pick voice, preview voice, engage. No separate "Characters" card was added — the consolidation kept the console grid tidy.

2. Structure

Files touched:

- `portal/server.py` — `api_encounter_voices_get` (route

`GET /api/encounter/{id}/voices)` now hydrates each selected persona id through `library.get_persona()` and returns: `- personas` — array of `{id, name, role}` triples - `join_code` — the encounter's join code, echoed so the client can build the Engage link without an extra round-trip - Backward-compat fields `selected_personas`, `patient_persona_id`, `voice_assignments` are preserved exactly.

- `portal/templates/encounter_console.html` — `<h2>` of `#card-voice`

renamed from `"■ Character voices"` to `"■ Characters · voices · engage"`; help paragraph rewritten to document the `■ Test` and `■ Engage` controls.

- `portal/static/encounter_console.js` — `bootVoices()` rebuilt:

reads the new `personas` array (with name + role); each row now has a `.char-label` block (name + role tag + faint pid), the existing `<select>`, and a `.char-actions` block with `■ Test` + `■ Engage`. New `testVoiceForRow(btn)` function handles the preview (POST `/api/tts` for ElevenLabs voices, browser SpeechSynthesis as fallback).

- `portal/static/encounter_console.css` — `.voice-row` regriddded to

three columns (label | select | actions) with mobile collapse to one column; new `.char-name`, `.char-role-tag`, `.char-role-tag.patient`, `.char-pid`, `.char-actions`, `.char-test`, `.char-engage` styles.

No schema migration. No new dataclass fields.

3. Uses

3.1 Operator flow

1. Instructor opens `/portal/room/encounter/{id}` (the Per-Patient Console for a single bed). 2. Voice card boots — `bootVoices()` fetches `/api/voices` (the ElevenLabs voice catalog) and `/api/encounter/{id}/voices` (the per-encounter persona roster + saved assignments). 3. Card renders one row per persona, ordered by `selected_personas`. The patient's row is tagged with a patient role badge; non- patient rows show their persona-library role (e.g. Anxious Spouse, Charge RN, Pediatric Patient). 4. Instructor picks a voice → change handler POSTs to `/api/encounter/{id}/voices` and writes `enc.voice_assignments` (same path as M30). 5. Instructor clicks **Test** → `testVoiceForRow()`: - If `voice_id` is set: POST `/api/tts` with the sample text + `voice_id`, gets back the audio stream, plays it via a transient `<audio>` element + `URL.revokeObjectURL` cleanup. - If `voice_id` is empty: uses `SpeechSynthesis` for an offline fallback preview. 6. Instructor clicks **Engage** → opens `/join?code={join_code}` in a new tab (`target="_blank" rel="noopener"`); they pick the target persona from the join flow's existing picker and chat as that character.

3.2 Data flow

```
GET /api/encounter/{id}/voices
  ■ library.get_persona(pid) per pid in enc.selected_personas
  ■ {personas: [{id,name,role}], join_code, voice_assignments, ...}
    ■ bootVoices() in encounter_console.js
      ■ row-per-persona render
        ■ select.change → POST /api/encounter/{id}/voices
        ■ ■ Test → POST /api/tts → <audio>.play()
        ■ ■ Engage → window.open('/join?code=...')
```

4. Functions (exported API surface)

Symbol	Where	Purpose
<code>GET /api/encounter/{id}/voices</code>	<code>portal/server.py</code>	Extended response: adds <code>personas: [{id,name,role}] + join_code</code> .
<code>library.get_persona(pid)</code>	<code>portal/library.py</code>	(existing) Used by the route to hydrate <code>id</code> → <code>{name, role}</code> .
<code>bootVoices()</code>	<code>portal/static/encounter_console.js</code>	Rebuilt — renders rows by display name + Test + Engage.
<code>testVoiceForRow(btn)</code> (new)	<code>portal/static/encounter_console.js</code>	Preview helper: POST <code>/api/tts</code> for ElevenLabs voice, falls back to browser <code>SpeechSynthesis</code> .

5. Limitations

- No persona-name search in the voice dropdown. With 12+

ElevenLabs voices and a row per persona, the dropdown can get long. The voice catalog itself isn't filtered by gender / age to match the persona — instructor judgment for now. A future M34 could auto-recommend voices based on the persona's `voice_profile` slot.

- **Engage button does not pre-select the persona.** The link

opens the generic `/join?code=...` page; the instructor still picks the target persona from the join-flow's dropdown. We considered adding `?persona={pid}` and a JS auto-select in `student_join.js`, but that touches the student-facing flow and was scoped out. Tracked as a follow-up.

- **No "Engage as patient" shortcut.** All rows get the same

Engage button; the patient row isn't special-cased. The role badge makes the patient obvious enough.

- **Voice test reuses `/api/tts` (public, no auth).** That route

was already station-facing (audio src tags need to fetch without cookies). The Test button benefits from that. If a future revision tightens `/api/tts` to require a session token, M33 will need a dedicated `/api/encounter/{id}/voice_preview` wrapper.

- **Preview sample text is hard-coded English.** Personas with

non-English voice profiles (none ship today) would need a localized greeting. Acceptable for v7.0.

- **Browser SpeechSynthesis voices vary by OS.** Fallback is

best-effort — the instructor will hear "something", but the voice characteristics (gender, accent) are whatever the browser default is, not a per-persona match.

6. Test status

Test file	Asserts	Status	Last run
<code>tests/v7/test_encounter_characters_voices_engage.py::test_voices_endpoint_returns_personas_with_name_and_role</code>	personas[] has id+name+role, join_code echoed, backward-compatible fields intact	PASS	2026-05-27
<code>...::test_voices_endpoint_defensive_when_persona_missing_from_library</code>	Unknown pid echoes id as name (no 500)	PASS	2026-05-27
<code>...::test_voice_card_h2_renamed_to_characters_voices_engage</code>	Console page contains the new card title	PASS	2026-05-27

Test file	Asserts	Status	Last run
...::test_encounter_console_js_renders_name_test_and_engage	JS source carries char-name, ■ Test, ■ Engage, testVoiceForRow, SpeechSynthesis, /api/tts, /join?code=markers	PASS	2026-05-27
...::test_engage_link_url_serves_join_page	The /join?code={join} URL the link targets serves 200/3xx (not 404)	PASS	2026-05-27
Full v7 suite	209 passed, 1 skipped (M20 Playwright skip — unchanged)	PASS	2026-05-27

7. Change list

Date	Author	Change	Files
2026-05-27	claude-code	Initial M33 implementation: voices route hydrates name+role+join_code; voice card renamed; JS rebuild with name labels + ■ Test + ■ Engage; CSS reflow; 5 new tests	portal/server.py, portal/templates/encounter_console.html, portal/static/encounter_console.{js,css}, tests/v7/test_encounter_characters_voices_engage.py (new)

8. Open questions / known issues

- **Engage button should ideally pre-select the persona.** Currently

the instructor must pick the target persona again on the join page. Tracked: add ?persona={pid} query-param honoring to portal/static/student_join.js so the join flow auto-selects.

- ****What happens if the operator clicks Engage during a private-clone**

encounter?** The join code is the template encounter's; the join flow will spawn another clone for that "instructor student". The cohort debrief would surface this as an extra student. We may want a flag to mark instructor sessions and exclude them from the debrief facet metrics. Out of scope for M33.

- **Test button rate limiting.** Spamming ■ Test would burn

ElevenLabs character budget. The per-encounter voice budget (M17) covers this — when the budget runs out, the route returns the M17 "fall back to browser TTS" error and the JS surfaces it in #voice-status. Verify in

LAN test.

- **Audio object cleanup.** `URL.revokeObjectURL(url)` fires on ended — if the instructor navigates away mid-playback, the blob URL leaks until tab close. Acceptable; the blob is small.

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